Ethan Yap

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PROFILE

Year 2 Information Technology student with a minor in Business, seeking a software engineering internship. Builds full-stack web apps with Node.js, Next.js and Python on PostgreSQL and MySQL, from REST APIs and real-time dashboards to AI search with embeddings, shipped with Docker and GitHub Actions. Personal rule: a working build seven days before every deadline, with a week left to test.

EDUCATION

Singapore Polytechnic · Diploma in Information Technology, Minor in Business — Apr 2025 – 2028 (expected)
GPA 3.49 / 4.00 (Year 1) · Modules: Back-End & Front-End Development, Java & OOP, Database Systems, Web & Mobile Apps

Maris Stella High School · GCE O-Level — 2021 – 2024
L1R5 net 10 · English Language A1, Mathematics A2, Additional Mathematics A2 · Direct School Admission (Robotics)

TECHNICAL SKILLS

Languages	JavaScript, TypeScript, Python, Java (incl. Swing), SQL, HTML5, CSS3 · **Spoken:** English, Mandarin
Backend	Node.js, Express, FastAPI, REST APIs, WebSockets, JWT/bcrypt auth, Zod validation, job queues, schedulers
Data & AI	PostgreSQL, pgvector, MySQL, SQLite, pandas, ETL, ChromaDB, RAG, Gemini API, sentiment analysis
Frontend	React, Next.js, Tailwind CSS, Bootstrap 5, Web Components, Chart.js, Recharts, responsive & accessible UI
DevOps & tools	Docker & Compose, nginx, GitHub Actions & Pages, pytest, Jest, Supertest, node:test, Git, Figma, Power BI

PROJECTS · live demos and code at ethanyap123.github.io/portfolio

Incident Monitoring Dashboard · *Docker Compose, Node.js, PostgreSQL, WebSockets, Next.js, GitHub Actions*

- Designed a five-container Docker Compose stack (simulator, Express collector, PostgreSQL, webhook receiver, Next.js UI) with health checks and start-up ordering, tested end to end in GitHub Actions.
- Wrote an alerting engine with hysteresis that opens, escalates and resolves incidents, sending webhooks with retries, and streamed live charts to the dashboard over WebSockets.

AI Code Snippet & Documentation Hub · *Next.js, TypeScript, PostgreSQL, pgvector, Gemini API, Zod*

- Built a job queue whose worker claims jobs safely across instances (SKIP LOCKED) to generate Zod-validated docs and tags.
- Stored 768-dimension embeddings in pgvector with an HNSW index for semantic search, re-ranked by Postgres full-text scores.

Academic Resource RAG Search Engine · *Python, FastAPI, ChromaDB, sentence embeddings, pytest*

- Built a PDF ingestion pipeline that tracks section headings and packs sentence-aligned chunks into ChromaDB.
- Grounded every answer in page-level citations, refusing off-topic and uncited answers; 26 pytest tests in CI.

Tech Trend Sentiment Tracker · *Python, pandas, SQLite, FastAPI, Next.js, Recharts*

- Built an ETL pipeline that pulls, cleans and de-duplicates headlines from eight sources every six hours on a GitHub Actions cron, scoring sentiment and tracking 11 sectors and 37 trend terms.

Pokémon Gym & Battle System · *Node.js, Express, MySQL, JWT, bcrypt*

- Built a full-stack game on an Express REST API over MySQL with JWT login, bcrypt hashing and ownership checks, a turn-based battle engine and a gacha shop run as single database transactions.

Student Internship Management System · *Java, OOP, Swing (pair project)*

- Wrote multi-field search with regex-validated IDs, and per-company headcount and score reports.

WORK EXPERIENCE

Two Men Bagel House · Line Chef (part-time) — Feb 2025 – Apr 2026

- Kept orders flowing between kitchen and front of house at peak; ran the register and barista station.

Bee Cheng Hiang · Barbecue Cook (part-time) — Nov 2024 – Feb 2025

- Ran a station alone through the festive peak, sequencing orders ahead of demand to shorten queues.

LEADERSHIP & COMMUNITY

Marist Ambassador · emceed 2023 prize-giving, presented at assemblies, guided Open Houses — 2021 – 2024

Marist Robotics & Programming Club · mentored juniors; Perse Coding Team Challenge (2022, 2023) — 2021 – 2024

Class Vice-Chairman (Sec 4) · **Peer Supporter** (Sec 2) — 2022, 2024

Community service · Mummy Yummy events (100+ participants), Magorium, Masjid Kassim — 2022 – 2025

AWARDS

Edusave Character Award, Ministry of Education and People's Association — 2024

Merit Award, Perse Coding Team Challenge (International) — 2023